

The Statistical Model

qsp-inference

One probability model, two questions: the single-patient posterior
and the virtual population

The section to read first. It sets down the probability model that the rest of the package implements, and the two questions we ask of it — no code, no Stage 1 / Stage 2, no jargon left undefined. Terms are given three names where they have three: one from Bayesian statistics, one from pharmacometrics, one from the simulation-based-inference literature, so that whichever field you arrived from, you can see the object is one you already know.

1 The setup

A mechanistic model of this kind carries a few hundred parameters — rate constants, half-saturation concentrations, homeostatic cell densities — and almost none of them can be measured in the clinical setting the model is meant to describe. The usual practice is to take a literature point value for each and attach a range wide enough to look cautious. That yields parameterizations that run, but it answers no question about how much the data actually constrain, and it cannot tell you when a calibration target is simply out of the model's reach.

We take the other route: write the model down as a probability model, and infer. One question runs underneath everything that follows, and it is worth stating plainly before the notation arrives.

If we trust the mechanism, and we trust the priors, then the calibration should fit the data on its own — whether the target is a single median patient or an entire population. Where it does, there is nothing left to hand-tune. Where it does not, the failure is telling us something, and the job is to say precisely *what kind* of failure it is.

Most of the machinery here exists to turn that question from a slogan into something you can actually answer.

2 The generative model

There is one model, and it has three pieces: a parameter vector, a prior over it, and a rule for how parameters produce data.

$$\theta \sim \pi(\theta), \quad x \mid \theta \sim p(x \mid \theta), \quad \mathbb{E}[x \mid \theta] = g(\theta).$$

The parameters $\theta \in \mathbb{R}^d$ live on a log scale wherever they are positive rates, so a prior is lognormal and the natural notion of distance between two values is a ratio, not a difference. The prior $\pi(\theta)$ is informative and built from external data; the next section is about how. The map $g(\theta)$ is the mechanism itself — integrate the ODE system forward and reduce each trajectory to the quantity that was measured — and it is deterministic: a given θ always produces the same $g(\theta)$.

The randomness is the measurement, and it is deliberately *not* assumed Gaussian. For each observable the noise is taken from the observed statistic's own bootstrap sampling distribution

— its skew, its tails, its bounds — and convolved onto the prediction, multiplicatively where the quantity is positive so that positivity is preserved. Only when a target supplies no bootstrap does a parametric fallback step in, and even then it is a lognormal or a Gaussian chosen *per observable* by the asymmetry of the reported confidence interval, not a blanket Normal. In the default path there is no parametric noise family at all: the observation layer is itself available only as a sampler.

The one feature that shapes every method downstream is that the likelihood $p(x | \theta)$ can be neither written down nor evaluated, for two compounding reasons: the mean map g is an ODE solve with no analytic form, and the noise layer, in its default, is an empirical resampler rather than a density. We can *draw* pairs (θ, x) as fast as we can simulate, but we cannot *evaluate* the density of one. Inference that needs only sampling — not evaluation — is called **simulation-based** (equivalently, *likelihood-free*): we learn the posterior with a neural density estimator trained on simulations, and the observation noise enters as augmentation on those training pairs, not as a likelihood we compute.

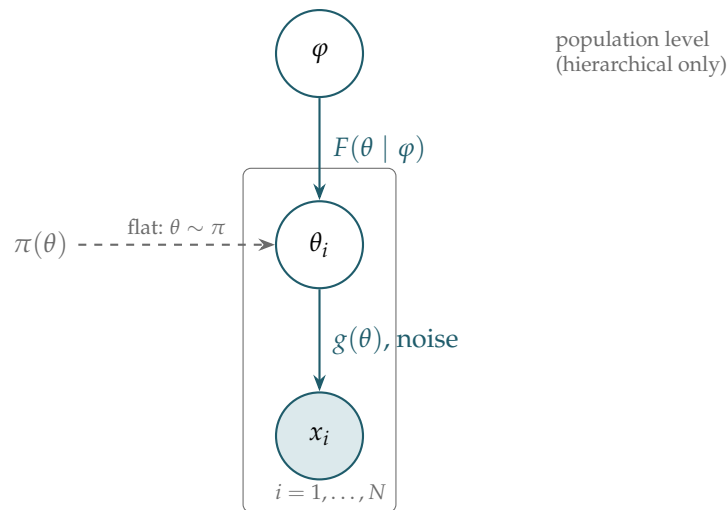


Figure 1: *

Cover the top node and it is the *flat* model, $\theta \sim \pi, x | \theta$. Include it and it is the *population* model: a shared φ generates each patient's θ_i .

3 Two questions, one model

The distinction between calibrating to a median patient and generating a virtual population is not a distinction between two workflows. It is two questions asked of the single model above, and it is the same distinction a mixed-effects modeler draws between a typical-value fit and a random-effects distribution.

The first question fixes one observed summary x_{obs} — the target medians — and asks for the posterior $p(\theta | x_{\text{obs}})$. This is calibration in the ordinary sense: one set of parameters, uncertainty and all, consistent with one observation. Call it the **flat**, or single-unit, or fixed-effects problem.

The second question does not treat the parameters as fixed across patients. It supposes each patient i has their own θ_i , drawn from a population distribution $F(\theta | \varphi)$ with hyperparameters $\varphi = (\mu, \omega)$ — a location and a spread — and the data are now summaries *across a cohort*: a median and an inter-quartile range for each target, over some real number of patients n_j . What we infer is φ , and the **virtual population** is the fitted distribution $F(\theta | \hat{\varphi})$. A virtual patient is

a draw from it. That is all a virtual patient is — a sample from an estimated random-effects distribution — and saying so is the whole point of writing the model this way.

	flat inference	virtual population
in Bayesian terms	single-unit posterior $p(\theta \mid x_{\text{obs}})$	hierarchical model; infer $\varphi = (\mu, \omega)$
in NLME terms	typical-value fit	the random-effects distribution (Θ, Ω)
the answer is	a posterior over θ	a distribution $F(\theta \mid \hat{\varphi})$ over patients
a “patient” is	a posterior draw	a draw $\theta \sim F(\theta \mid \hat{\varphi})$
the data are	target medians	target (median, IQR) across n_j patients

Nothing about the mechanism g , the prior π , or the observation model changes between the two. Only the question does. The package is arranged around that fact: the same simulator and the same prior serve both, and the population problem is the flat problem with a level added on top.

4 The prior is a measurement model, not a guess

For a model this over-parameterized the prior is not a nuisance to be set wide and forgotten. As the next section makes precise, the data identify only a handful of directions in parameter space; along the rest, *the prior is the answer the calibration returns*. A prior that carries the answer for two hundred parameters has to be built with the care one would give a likelihood.

So it is built, not asserted. Each quantitative claim in the literature is paired with a small forward model that predicts *that measurement* from a few parameters, and the pair is inverted — an ordinary measurement-error model. Three features are worth naming, because each corresponds to a standard statistical device.

A source measured in the wrong context should count for less. Each source carries a **translation sigma** that inflates its likelihood variance according to how far its setting — species, indication, assay — sits from the one being modeled. This is the between-study heterogeneity of a **random-effects meta-analysis**, the τ^2 that lets mouse data inform a human parameter without pretending to be human data.

It is also, exactly, a **power prior** (Ibrahim & Chen 2000) — the standard device for borrowing strength from historical data under a discount. A power prior raises a historical dataset’s likelihood to a fractional power, $\pi(\theta \mid D_0, a_0) \propto L(\theta \mid D_0)^{a_0} \pi_0(\theta)$, and for a Gaussian likelihood that is *identical* to inflating the variance:

$$L^a \propto \exp\left(-\frac{a(y-g(\theta))^2}{2\sigma^2}\right) = \text{kernel of } \mathcal{N}(y; g(\theta), \sigma^2/a).$$

So discounting a source by inflating its σ is discounting its likelihood by an exponent $a_j = \sigma_j^2/\tilde{\sigma}_j^2$; the two constructions coincide here and diverge only for non-Gaussian likelihoods. Worth stating plainly because it locates the relevance rubric in a literature with its own machinery: the rubric is a **fixed, hand-specified** choice of those exponents, while the *normalized* power prior treats a_0 as random and infers it from how consistent a source turns out to be with the others. That is the natural route to retiring the rubric as a manual surface, and it is affordable exactly here — Stage 1 chunks are small and already sampled with NUTS, so the normalizing constant that makes inferred a_0 expensive is within reach.

Parameters informed by the same measurement are correlated, and discarding that would overstate what we know. The per-parameter marginals are therefore joined by a **Gaussian copula** — Sklar’s construction, marginals for the uncertainty and a correlation matrix for the dependence — giving a genuine joint distribution that can be both sampled and evaluated.

A parameter with no data of its own, but a mechanistic tie to one that does, enters as a log-linear-Gaussian **derived** child of its anchor: it inherits the parent’s width and correlation instead of receiving an invented target of its own.

5 Inference without a likelihood

Since the likelihood can only be sampled, we estimate the posterior directly. Simulate many pairs (θ, x) from the model, fit a normalizing-flow conditional density $q(\theta | x)$ to them, and read it at $x = x_{\text{obs}}$. This is **neural posterior estimation** (NPE), the amortized member of the simulation-based-inference family: the cost is paid once, at training, after which any observation can be conditioned on cheaply.

Three refinements matter, and each is a named technique rather than a tuning knob. The distribution we *draw training samples from* need not be the prior we *report under*. We draw from a wider **proposal** $\tilde{\pi}$, wide in the directions the data can actually inform, where coverage is worth paying for, and recover the posterior under the anchored prior π by self-normalized importance weighting, $w \propto \pi / \tilde{\pi}$, watching the effective sample size. Separating the two is what lets the proposal be a computational choice and the prior a scientific one. Left to itself the proposal is still an ad hoc width; **truncated sequential NPE** (Deistler et al. 2022) removes that by narrowing it, round by round, toward the region the posterior occupies, which also keeps the importance weights well behaved. The result reported is the posterior under π ; the proposal has done its work and drops out.

6 Populations: the level on top

The population problem adds the hierarchy the flat problem omits. Patient i draws $\theta_i \sim F(\theta | \varphi)$, the hyperparameter $\varphi = (\mu, \omega)$ has its own hyperprior, and the observations are cohort summaries. We infer $p(\varphi | \text{summaries})$ by amortized inference over simulated cohorts, and emit the virtual population as draws at $\hat{\varphi}$.

Two choices here are statistical rather than incidental. First: you cannot estimate two hundred population spreads when only a few directions carry signal, so $\log \omega$ is inferred along the leading eigen-directions of π ’s covariance — the directions the data speak to — and left at its anchored prior value everywhere else. The identifiability limit that governs the flat posterior governs the population spread in exactly the same way, and the eigenbasis is where that limit is imposed. Second: the amount a target says about the spread should match how many patients it was measured in. A target reported over six patients constrains ω weakly; one over nine hundred constrains it tightly. Cohort summaries are drawn at each target’s real published n , so the evidence is weighted honestly.

An older construction — generate a cloud of plausible patients and reweight it to match the observed marginals (Allen et al. 2016) — remains available as a fixed-cloud fallback. The hierarchical model is preferred because it *infers* the spread from data instead of inheriting it from whatever width the proposal happened to have.

7 The hard part: most parameters are not identified

Everything above would be routine if the data determined the parameters. They do not, and it is better to confront this directly than to let it surface as mysterious posterior widths.

Only a handful of the model’s parameters are identified by the clinical data. Along every other direction the likelihood is nearly flat: many different θ produce almost the same x , so

the data cannot choose among them. This is **practical non-identifiability** — not a defect of the sampler, and not something more simulations would cure. It is the geometry of the model, the **sloppy** structure (Gutenkunst et al. 2007) that mechanistic models of this size almost always have: a few stiff directions hold all the constraint, and the rest are soft.

The consequence, stated without softening

For an unidentified parameter the posterior equals the prior, and the population spread equals the prior spread. The data have added nothing, and the honest report says so — no shrinkage on a soft direction is the correct answer, not a failure to converge. Reporting a confidently narrow posterior there would be reporting the proposal, not the data.

This is also why the prior work earns the name *science* rather than *tuning*. Along the identified directions the data speak and the prior yields to them. Along the soft directions the prior is the answer, and it can only come from outside the calibration — from the measurement-model priors and their derived children. No sampler, however elaborate, invents information the data do not contain. That external half of the model is irreducible, and it is the one part of this pipeline where a human is meant to be in the loop.

8 Checking the fit

A calibration is not to be believed until it has been checked, and each check earns its place by carrying a null distribution computed from the model’s own simulations — so that “this target is a poor fit” is a statement with a false-positive rate, not an impression. Taken together they are the diagnostics of an ordinary Bayesian workflow (Gelman et al. 2020), run in a fixed order because each step’s verdict is only trustworthy once the previous one has passed.

The gate is **simulation-based calibration** (Talts et al. 2018): draw a parameter from the prior, simulate, infer, and check that the true value lands uniformly within its own posterior. It never sees the real observation, so it validates the inference procedure rather than the model — but if it fails, no later verdict can be trusted, because the machinery itself is miscalibrated. With that passed, a **prior-predictive check** (Box 1980; Evans & Moshonov 2006) asks whether the model, run under the prior, produces the observed value at all, or whether x_{obs} sits in the tail where the anchored model rarely goes. When it does sit in the tail, one more question separates the two things that could be wrong: point the sampler at x_{obs} and see whether *any* admissible θ reaches it. If some does, the mechanism is fine and the prior is merely aimed elsewhere — a matter of re-anchoring. If none does — if the inverse map’s own best guesses, simulated forward, still miss — then the mechanism cannot produce the observation, and no prior will rescue it. A final **joint check** (a Mahalanobis discrepancy against a self-reference null, with per-observable leave-one-out influence) catches the case where every target is individually reachable but no single θ reconciles all of them at once, which points not at any one observable but at the coupling between them.

It is worth being clear about what this suite does and does not do, because the failure mode it exists to prevent is a seductive one. It *localizes* a miss and it *gates* the next step; it does not prescribe the fix. No scalar summary can tell you which mechanism to add or which parameter value is right — that reading always comes from a mechanistic trace and knowledge of the biology. A diagnostic that hands you a parameter to turn is not saving you work; it is guessing, and a calibration tuned to a guessing diagnostic will confirm whatever it was pointed at. The suite’s job is to say that a miss is real, where it lives, and whether re-anchoring or a mechanism change is called for — then to hand a short, ranked list to the trace, where the diagnosis is

actually made.

9 A dictionary

Every term this package uses has an older name, usually two. The map from one vocabulary to the others:

here	Bayesian statistics	pharmacometrics / NLME	ref.
SubmodelTarget → Stage-1 posterior translation sigma	measurement-error / meta-analytic prior between-study heterogeneity τ^2 ; equivalently a power-prior discount a_0	literature-informed prior inter-study variability	Box & Tiao 1973 DerSimonian & Laird 1986; Ibrahim & Chen 2000
copula prior derived prior flat inference virtual population / patient (μ, ω) prevalence weighting NPE / SBI	marginals joined by a copula log-linear-Gaussian child single-unit posterior random-effects distribution / a draw population hyperparameters importance reweighting to fixed marginals amortized, likelihood-free posterior	correlated random effects structural/covariate relation typical-value fit population distribution / subject Θ, Ω VPop selection	Sklar 1959 — — Gelman & Hill 2007 — Allen et al. 2016 Cranmer et al. 2020
TSNPE	truncated sequential NPE	—	Deistler et al. 2022
proposal → prior reweight identifiability wall	self-normalized importance sampling practical non-identifiability	— —	Kong et al. 1994 Gutenkunst et al. 2007
eigenbasis reparameterization SBC prior-data conflict self-reference D^2 / LOO influence LOO-PIT (<i>planned</i>)	identified subspace / stiff directions simulation-based calibration prior-predictive check predictive discrepancy + leave-one-out leave-one-out probability integral transform	— — — — —	Gutenkunst et al. 2007 Talts et al. 2018 Box 1980 Vehtari et al. 2017 Gelfand et al. 1992
misspecification detector OBED	model criticism for amortized inference Bayesian experimental design	— optimal design	Schmitt et al. 2023 Chaloner & Verdinelli 1995

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